Danny Muir

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Education

University of Washington, Seattle — *Expected June 2027*

Bachelor of Science in Computer Engineering, Paul G. Allen School of Computer Science & Engineering — **GPA: 3.84**

Minor: Neural Computation and Engineering — **Dean's List:** All Quarters 2023–2026

Honors: Tau Beta Pi Engineering Honor Society; CSE Honors

Relevant Coursework: Machine Learning, Neural Technology Studio, Neurocomputation/Engineering Lab, Natural Language Processing, Scientific Computing, Linear Algebra, Data Structures & Algorithms, Hardware/Software Interface

Experience

Undergraduate Researcher, Neural Systems Laboratory, UW — *Oct 2025–Present*

Advisor: Dr. Rajesh Rao, Paul G. Allen School of Computer Science & Engineering

- Optimizing multi-session neural decoding framework in collaboration with Matthew Bryan, building on temporal basis function models (TBFMs) for closed-loop neural stimulation.
- Testing and selecting optimal implementations for session-specific embeddings and test-time adaptation methods.
- Improving training efficiency, model complexity, and cross-session generalization through systematic empirical evaluation.
- Conducting honors thesis research on learning dynamical systems underlying traveling waves in motor cortex to forecast motor-relevant neural subspaces.

Associate Researcher, VP Operations, Sensors, Energy, and Automation Laboratory (SEAL), UW — *Jan 2025–Oct 2025*

Advisor: Dr. Alexander Mamishev, Department of Electrical & Computer Engineering

- Conducted VLF/ULF signal processing research for atmospheric and geophysical monitoring applications.
- Contributed written sections and managed submission logistics for federal and industry grant proposals (DOE STTR, NIH, NSF, CDC, Google).
- Led project management team developing custom automation and DevOps solutions for laboratory operations.

Leadership

President, Triangle Fraternity, Washington Chapter — *Nov 2025–Present*

- Established Triangle UW's first Interfraternity Council membership; scaled sorority collaborations from ~1/quarter to 11 events across two quarters with 4 partner chapters; grew chapter from 21 to 40 members.
- Rebuilt executive board from 6 to 8+ formalized roles — adding Housing and Communications Directors and formalizing Academic Chair and Rush Chairs — distributing operational ownership and building institutional continuity.

Vice President & New Member Educator, Triangle Fraternity, Washington Chapter — *Dec 2023–Nov 2025*

- Built and systematized chapter event operations, enabling consistent execution of social and brotherhood programming.
- Served as chapter academic lead; chapter earned the Kahlert Academic Excellence Award (highest GPA of any Triangle chapter nationally) and Most Improved Chapter recognition from national Triangle.

Skills

Programming: Python, Java, C, JavaScript, Bash, Git, HTML, LaTeX | **Platforms:** macOS, Linux, Windows

ML/AI: PyTorch, NumPy, Pandas, scikit-learn, MAML/Meta-Learning, Test-Time Adaptation, Autoencoders, Neural Networks, Computer Vision, Transformers

Neuroscience: ECoG/cortical probe signal processing, sEMG, neural decoding, temporal basis function models, closed-loop BCIs

Research: Grant writing, experimental design, project management, scientific computing, technical documentation